A SHORT HISTORY · 1854–1866

A Line That Could Hold

The Atlantic cable, from failed attempts to a working connection

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Chapters, exact source excerpts, and an offline evidence ledger

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Preface — The difference between arrival and connection

In late July 1866, a message from Newfoundland announced that the Atlantic cable had been laid and was working. Henry M. Field reproduced the announcement in his history of the undertaking: his brother Cyrus reported reaching Heart's Content on July 27 at nine in the morning. Yet Henry also recorded a delay. A broken cable across the Gulf of St. Lawrence prevented the news from reaching New York until July 29. The ocean had been crossed, but the message still needed a way onward. This small interruption supplies the starting point for our story. [1]

> **Interpretation.** The useful question is not simply who first stretched a wire across the Atlantic. It is how a connection that could briefly demonstrate its possibility became one that people could begin to rely upon. The difference joins several histories: manufacturing and seamanship, electrical measurement and financial confidence, public celebration and daily maintenance. The completed span mattered enormously, but its endpoints were not the endpoints of every message. A history attentive to the unfinished journey can appreciate the achievement without treating a landing ceremony as the end of the work. [1] [2]

This short book follows the attempts that culminated in the connections of 1866. Its principal witness for the 1865 voyage is William Howard Russell, whose published account also looks back at the earlier expeditions. For 1866 it uses Cyrus Field's banquet speech, a selected page from Henry Field's contemporary history, and two newspaper items printed in Newfoundland on August 1. The bibliography explains the surviving digital versions. These are real historical texts, but their evidence has limits: a participant's recollection, a reproduced telegram, and a newspaper report are different kinds of testimony.

The account does not invent conversations, private thoughts, or scenes to fill those limits. Paragraphs marked “Interpretation” state the author's reading of the evidence, rather than an additional fact established by a quotation. Numbered notes lead to exact excerpts and source locations; the accompanying evidence edition makes the full retained text available offline. Where a source makes a prediction or defends its author's conduct, the narrative keeps that distinction visible. This is a bounded history through 1866, not a measurement of the cable's long-term economic effects or a comprehensive history of everybody whose labor made it possible.

Chapter 1 — Before the ocean

The Atlantic enterprise grew out of a more limited effort to improve communication through Newfoundland. Russell describes F. N. Gisborne arriving in New York early in 1854 after his company's financial difficulties and suspension of work. There he obtained an introduction to Cyrus Field and presented his plans. This is an origin with a failed undertaking already inside it. The prospective ocean connection did not begin in an empty landscape: it emerged from an existing project that needed money, from rights already sought, and from work that could not continue on its previous footing. [3]

Field's own account of the early overland work is revealing. Addressing the New York Chamber of Commerce in November 1866, he recalled how easy it had been to draw a line on a map. The map concealed forests, mountains, swamps, rivers, and gulfs. He and his associates had not seen the country and expected construction to take a few months; he said it took two and a half years. The speech celebrates persistence, but the admission also records an initial failure to understand the task they were financing. [4]

The project depended on political permissions as well as private enthusiasm. Russell says the Newfoundland legislature granted privileges which amounted to a monopoly

of telegraphic rights there. His account presents these concessions as an encouragement to risk capital, not merely an administrative detail. A cable required somewhere to arrive, and rights at that arrival point had value before the ocean span existed. The promised connection was therefore already an institutional arrangement as well as a proposed machine. The evidence here is Russell's description of those privileges; the underlying legislation has not been independently examined for this book. [5]

Even the shorter water crossing proved difficult. In August 1855, according to Russell, an attempt to lay a cable across the Gulf of St. Lawrence met a violent gale off Cape Ray. The vessel's master considered it necessary to cut the cable. A further contract placed the risk of making and laying another line with the manufacturers, and Canning carried out that undertaking in the Propontis the following year. Long before the best-known Atlantic disappointments, the connected route had already required a second attempt. [6]

The Atlantic itself first had to become an object of investigation. Russell records that Lieutenant Berryman, working from the American steamship Arctic in July 1856, took soundings at intervals across the ocean and brought up specimens from the bottom. Microscopic examination identified fine shells and sand. These observations did not amount to a continuous view of the seabed. They were samples used to make a project on an immense and largely

unseen route more thinkable. The distinction between a measured point and the space between points would remain important when cable later disappeared below the ships. [7]

> **Interpretation.** The work on land, the failed Gulf crossing, and the soundings suggest three different meanings of preparation. One was arranging authority and money. Another was building actual links in a route. A third was turning uncertainty into observations on which an engineer might act. None could substitute for the others. A favorable opinion about the ocean floor could not clear a Newfoundland road; a charter could not prevent a cable from breaking. This helps explain why a persuasive plan could coexist with expensive and repeated practical failure. [4] [5] [6] [7]

By December 1856, preparation had become a manufacturing order. Russell describes agreements for 2,500 miles of insulated copper core, with the finished cable divided between Glass, Elliot & Company and Newall & Company. He gives July 6, 1857, as the date by which the whole cable had been completed. The figures belong to his account of the order, rather than to a new audit of the manufacturers' books. They nevertheless convey the change in scale: a speculation discussed in offices now existed as a long, heavy object that ships would have to manage. [8]

Interpretation. The relevant achievement at this stage was not that the promoters had proved the whole route workable. It was that they had assembled enough resources to attempt it. To read the order backward from the success of 1866 would hide the uncertainty of 1857. The manufactured cable was both an investment and a question. Its passage through machinery, across a moving ship's stern, and down to the seabed would test arrangements whose performance could not be established merely by completing the order on time. [8] [9]

Chapter 2 — The sea takes the first answers

The first Atlantic attempt ended with the loss of the line. Russell places the decisive break on August 11, 1857, after about 330 nautical miles of cable had been laid. The Niagara was roughly 280 miles from Valentia. His account gives a speed of three to four knots and a faster rate of cable payout; the cable itself was not a taut straight line equal in length to the distance traveled by the ship. Managing the excess while controlling strain was part of the work, not an incidental complication. [9]

Russell reports that the strain indicator showed 3,000 pounds, while adding that the actual strain was no doubt greater. That qualification matters. A number displayed by an instrument and the force experienced by the cable were not automatically identical descriptions of the event. The account does not allow us to reconstruct every motion at the moment of failure. It does allow us to see the engineers confronting a system in which ship movement, machinery, and a long submerged weight interacted faster and less conveniently than a simple diagram might suggest. [9]

There was also a limit to what could be attempted after the break. Russell says the officers and engineers judged another immediate effort unwise with only 1,847 miles of cable

remaining aboard the ships, which he describes as an allowance of twelve percent above the distance required. The decision was not simply a verdict that the idea was impossible. It concerned the material available for the next attempt and the margin it provided. A broken cable consumed both money already spent and the physical reserve with which another uncertainty could be faced. [10]

The machinery used for the next campaign embodied a response. Russell contrasts the earlier brake, which a hand movement could make exert very great resistance, with a new arrangement that allowed the operator to ease the cable quickly. An officer watched the strain register and could release resistance as the ship's stern rose. In Russell's explanation, that officer could not increase the strain beyond the setting established by the engineer. Responsibility was being divided between an initial adjustment and the moment-to-moment handling of the cable. [11]

The plan for 1858 also changed the geography of the operation. Instead of beginning with a shore end at Valentia and proceeding outward, the cable ships were to meet midway across the Atlantic, splice their portions together, and steam toward opposite shores. The Agamemnon would carry one end toward Ireland and the Niagara the other toward Newfoundland. In this arrangement neither ship alone had to complete the full crossing while paying out the entire line. The success of the whole still depended on both portions remaining connected. [12] [13]

The June expedition did not enjoy the orderly conditions that such a plan suggests on paper. Russell records the ships leaving Plymouth on June 10 after satisfactory preliminary trials, then encountering a prolonged storm. His account describes the Agamemnon rolling heavily while trying to reach the rendezvous. A successful test cruise therefore preceded, rather than prevented, a severe encounter with conditions at sea. The loaded vessel and its cable had to survive that journey before the central operation could even begin. [14]

At the rendezvous, repeated efforts failed. Russell reports a first splice on June 26, followed by a break when the cable fouled a scraper on the Niagara. After another splice the electrical continuity ceased, and the Niagara's cable also snapped while being hauled in. A third effort brought another loss. In that case Russell stresses how moderate the speed and strain appeared and how favorable the weather was. His inability to supply a satisfying explanation is itself part of the record: not every setback arrived with a clear lesson attached. [15]

> **Interpretation.** Failure should not be romanticized into automatic progress. Changing a brake or a route could address a known problem, but an unexplained break remained unexplained. The useful distinction is between persistence and repetition. Persistence could include revising equipment, preserving reserves, and admitting uncertainty. Repetition alone would have meant sending

> the same arrangement out again and expecting desire to alter its behavior. The evidence from 1857 and June 1858 shows specific changes, but it gives no warrant for saying that each failure was necessary or that success was inevitable. [11] [12] [15]

Chapter 3 — A connection that could not last

The ships tried again in late July. Russell places their meeting in mid-ocean at midnight on July 28, 1858, and the splice on the following morning. His narrative then follows the accumulating length of cable paid out. On August 5, the Agamemnon was at Valentia and the Niagara at Trinity Bay, Newfoundland. The latter received a signal that the cable had been landed from the Agamemnon. For the first time in this sequence of attempts, both ends had reached their destinations with communication between them. [13]

The celebratory messages offered a language for the event. Queen Victoria's message to the American president congratulated him on the successful completion of the international work and expressed the hope that the cable would strengthen a friendship based on common interests and reciprocal esteem. Russell reproduces that message. Its importance is not that it proved the cable would secure peace. It shows what a sovereign wished the new connection to mean, and how diplomacy could turn a technical achievement into a statement about relations between nations. [16]

There is evidence of actual traffic, although the record used here does not justify treating every transmission as a commercial customer's message. Russell's later summary gives 271 messages from Newfoundland to Valentia and 129 in the opposite direction during 1858. The total is four hundred. That supports the modest but essential conclusion that the connection carried messages. It does not establish a sustained, profitable public service, nor tell us how the individual transmissions were classified. Those questions would require the traffic records behind the published total. [17]

By early September, the problem was public. Russell reproduces a letter dated September 6 from the company's secretary, George Saward. It said that no intelligible signals had been received from Newfoundland since one o'clock on Friday, September 3. The directors were investigating a fault whose location had not yet been discovered. No date could be given for opening the wire to the public. The document is narrower and more useful than a dramatic claim that the cable simply died at a single universally agreed instant: it describes a particular direction of signaling and a particular state of knowledge. [18]

Russell's discussion of the failure contains several explanations, with different degrees of confidence. He calls an account involving insufficient slack over a steep seabed slope conjectural. He thinks attempts to overcome defective insulation with increased battery power probably hastened

the breakdown. He also reports a committee's conclusions about hasty manufacture, unequal mechanical strain, and the injury caused by repeated coiling and uncoiling. These are not interchangeable causes. Nor is his summary of an inquiry the same thing as the full inquiry's evidence. [19]

This book therefore does not reduce the failure to one person's recklessness or one definitive voltage figure. The retained evidence supports a cable already vulnerable to damage and a contemporary argument that the way it was worked aggravated its condition. It does not support a complete allocation of individual blame. Keeping that limit visible does not make the technical failure less important. It prevents the convenient simplicity of a villain from replacing the harder question of how manufacturing, handling, and electrical operation fitted together. [19]

The material did not cease to be instructive when communication ceased. Russell records a recovery effort from Newfoundland in 1860 that brought up five miles of cable. He says the gutta-percha insulation was found undeteriorated and the core's electrical condition improved by submersion. Later attempts on the Irish side in 1862 achieved no practical advantage and were abandoned amid storms. These observations offered a reason to distinguish the fate of a particular installed line from a blanket judgment that deep seawater necessarily destroyed the insulating material. [20]

Interpretation. The 1858 cable both succeeded and failed, depending on the question asked of it. It demonstrated that signals could cross the Atlantic through an installed cable; it did not establish the dependable service its sponsors wanted. Those statements should be allowed to coexist. Celebration had compressed them into one achievement, while disappointment could reverse the error and deny the demonstration altogether. A useful history keeps the narrower success intact and records the larger failure without disguising either. [17] [18]

Chapter 4 — Building a different attempt

A further expedition required more than an argument that the idea remained possible. Field's banquet speech recalls a stage when the promoters needed £600,000 but had raised less than half. He described the enterprise as at a deadlock and needing help from another quarter. This is testimony from the man trying to obtain that help, delivered after the eventual success. It should not be mistaken for an independent account of every investor's reasoning. It does show that conviction and the money needed to act on it were separate resources. [21]

The new cable also differed materially from its predecessor. In the manufacturing account reproduced by Russell, the copper conductor weighed 300 pounds per nautical mile, compared with 107 in the 1857 cable. The insulating material weighed 400 pounds per nautical mile, compared with 261 for the earlier cable. These figures describe specified construction, not a new experiment performed for this book. Their significance here is that the renewed attempt did not merely remanufacture the old object and ask a larger ship to carry it. [22] [23]

Testing was incorporated into the description of production. Russell's account says the completed core was immersed for twenty-four hours in water at 75 degrees Fahrenheit before tests of conductivity and insulation. It explains that heat reduced gutta-percha's insulating ability, so the conditions were deliberately unfavorable. The aim was not to find the most reassuring possible reading. It was to test under a stated condition that could expose weakness and permit comparison. That account remains a manufacturer's testing narrative, rather than independent confirmation of every length delivered. [24]

The Great Eastern gave the enterprise a different vessel around which to organize its work. Field recalled the cable being taken aboard as it became ready and coiled into three large tanks. His speech dates the ship's departure on its 1865 voyage to July 15. This was not yet the successful ocean connection of the following year. In the story told after success, the enormous tanks could appear as preparations for an inevitable triumph; in the actual sequence they carried a cable which would be lost before reaching Newfoundland. [25] [26]

Russell identifies two groups of electricians aboard during the laying operation. The Telegraph Construction and Maintenance Company's electricians, under de Sauty, tested the cable and maintained signals with shore. Varley, Thomson, and assistants represented the Atlantic Telegraph Company. Their role was to observe and report on the

testing, and ultimately to certify whether the delivered cable met the contract. Russell explicitly says that this second group did not have power to interfere with or control the operation. Expertise did not erase the boundaries of contractual authority. [27]

That distinction would matter when something went wrong. A cable might be able to pass messages without meeting the promised standard; an electrician might believe it usable without having the authority to order the ship onward. Russell later reports precisely this tension in the discussion of the final fault. For now it is enough to recognize that “the expedition” was not one mind with one decision rule. Its participants shared an objective while also performing different jobs for organizations with different responsibilities. [27] [28]

> **Interpretation.** The new attempt combined material change with a more explicit means of judging performance. More copper or insulation alone would not tell a purchaser whether the finished line met its obligations. A test alone would not carry the cable over the side safely. And a favorable judgment by a respected scientist did not necessarily authorize a departure from the contract. These distinctions can sound bureaucratic beside the drama of a ship at sea, but they help explain why the practical decisions were harder than simply choosing between courage and caution. [22] [23] [24] [27]

The reader should also notice how much of this account comes from interested participants. Russell prints a detailed description of manufacture and gives strong attention to the men conducting and supervising it. Such material is valuable precisely because it preserves the procedures and expectations of those responsible. Its value does not require us to accept every assurance of perfection. The next voyage would bring the carefully prepared cable into circumstances in which specified strength, observed electrical behavior, and the handling of machinery all had to work together. [24] [27] [26]

Chapter 5 — What the lost cable taught, and what it did not

An early interruption in 1865 revealed an unexpectedly intimate source of damage. In his entry for July 25, Russell describes a faulty portion hauled back aboard. A short piece of wire, of the kind used in the cable's own protective strands, had penetrated its covering and injured the insulation. He records the initial view that wire had accidentally entered the tank and been forced into the cable as it passed through the paying-out machinery. The account identifies a physical defect while leaving its route into the cable uncertain. [29]

Later discoveries complicated suspicions that someone had deliberately damaged the line. Russell describes a further piece of brittle wire and reports the resulting division between those who believed in malice and those who thought the damage accidental. By his account, nearly everyone eventually favored an accidental explanation. He came to regard the cable's intended protection as a possible source of its weakness. This is evidence of changing judgments aboard the ship, not proof that every earlier suspicion was justified or that the precise history of each fragment could be reconstructed. [30]

The final fault also produced disagreement about the right course. Russell reports Thomson's later opinion that messages could have continued through the fault for a considerable time, and de Sauty's belief that the cable might have worked for months. But he adds that Canning had no reason to act on those views and that Varley could not have certified the cable to the contract standard at Newfoundland. A choice that looks obvious after a disaster may have looked different when the immediate obligation was to deliver a tested, compliant line. [28]

On August 2, the cable was lost during the attempt to haul it in. Russell's description follows a damaged portion approaching the ship, a sudden jar, and an indicator jumping beyond its marked scale. The shackle and wire rope came out of the groove of a wheel and fell onto the neighboring smaller wheel. The cable then parted and disappeared into the sea. Russell leaves the source of the initiating jerk uncertain, suggesting either a heave of the vessel or the secured shackle. The narrative is specific without being a complete mechanical reconstruction. [26]

The loss did not end the effort at once. Grappling sought a line too deep to see directly. In his retrospective summary of the voyage, Russell counts four attempts and says that the cable was caught in three of them. Catching it was not equivalent to bringing it aboard, repairing it, and completing a transatlantic connection. That distinction is central to the

result of 1865. The attempt had failed at its main task, while the attempts to retrieve the cable supplied grounds for considering a future recovery possible. [31] [32]

The directors made that possibility part of their next plan. Their September 1865 report, reproduced in Russell's appendix, described a contract for a new cable to be made and laid in 1866. The contractors were also to take sufficient cable and equipment to use their best endeavors to recover and complete the broken one. Profit on the new cable was contingent upon success. Recovery was thus more than a consoling suggestion after a loss: it became an obligation to attempt a defined piece of work alongside the next laying expedition. [32]

Russell's confidence at the end of his book went further than the observations alone. He described alterations to the Great Eastern that would make her “absolutely perfect” for laying the new cable and picking up the old. Read in its place, the sentence is a prediction written before the 1866 outcome. It belongs to the atmosphere of renewed expectation. It is not an assessment that this history can adopt literally, and the eventual success cannot retroactively turn that prediction into proof that no risk remained. [33]

> **Interpretation.** The failed expedition produced evidence of capabilities as well as of vulnerabilities. A fault could be found and brought back; a cable lying deep in the ocean could be caught; a particular recovery system could still

> fail. The useful lesson lay in keeping those findings separate. It would be misleading to call 1865 a completed connection because it made 1866 more plausible. It would be equally misleading to regard all its observations as worthless because the principal undertaking had failed. [29] [31] [26]

Chapter 6 — The connection, and the second cable

Before the next departure, the project met a difficulty that lay outside the ocean. Field recalled that the English attorney-general had advised that the proposed new twelve-percent shares could not be issued without a special act of Parliament. Work stopped and subscriptions were returned. In his account, the response was to create the Anglo-American Telegraph Company. The episode belongs to his retrospective speech, but its place in the sequence is clear: the ability to finance the next attempt required an organizational change, not simply another improvement to the cable. [34]

Field said that the public subscription, opened through J. S. Morgan & Company, raised the required £600,000 in fourteen days. That statement records his claim about the speed and completeness of the financing. It does not tell us, by itself, what each subscriber knew or how the investment was valued. The important practical contrast with the earlier deadlock is that enough money was now said to be available to proceed. The planned expedition had moved from an expression of confidence to a funded undertaking. [35] [21]

In describing the crossing, Field resisted an explanation based on unusually favorable weather. He said a dispatch he had written as “unpleasant” had appeared in New York newspapers as “pleasant,” and recalled fogs and storms on the passage. His explanation stressed science combined with practical experience. This book has not assembled a continuous weather log to test that judgment. What the speech securely shows is the explanation he wanted his audience to accept: organization and experience, rather than a fortunate interval of calm, deserved the credit. [36]

The expedition reached Newfoundland on July 27, 1866. For that date and the report of arrival, this book uses the selected page of Henry Field's history, which reproduces Cyrus Field's announcement from Heart's Content. It reports the cable laid and working. Henry's narrative then preserves the awkward sequel: the broken Gulf of St. Lawrence cable delayed the news to New York until July 29. A successful Atlantic crossing and a delayed end-to-end message were not contradictory events. They were simultaneous features of the same emerging route. [1]

A Newfoundland newspaper provides a local check on that sequel. The Times and General Commercial Gazette of August 1 printed a Heart's Content report dated July 29. It described the Bloodhound leaving with new cable aboard for repair of the Gulf line. It also said that the Great Eastern would remain that week before going to pick up the old

Atlantic cable. Here the immediate aftermath appears as movement and preparation: ships carrying material, work still required, and a recovery expedition still ahead. [37]

The same newspaper printed a further telegram dated July 31, signed Cyrus W. Field. It said the Queen's message had been delivered to the president and a reply sent through the cable to the Queen; the Great Eastern was firing a salute. The newspaper is an independent surviving publication, but the telegram is still a communication attributed to Field. It corroborates contemporary reporting of actual use without becoming an independent technical test of every part of the system. Its value lies in bringing us close to what readers were told while the event was still new. [38]

Recovery of the 1865 cable demanded a return to the place of loss. Field described marking its line with numbered buoys, with flags by day and lanterns by night, and drifting across the route with a grapnel dragging the bottom. In his account, a cable brought into sight on August 17 escaped again. The successful effort caught it on the last night of August and brought it aboard after midnight on Sunday morning. The speech makes clear how a moment of apparent recovery could still turn into further work. [39] [40]

Once aboard, the recovered cable was taken to the electricians' room. Field recalled the suspense before a signal established that it could communicate again. The line was then completed to Newfoundland in early September,

giving the undertaking a second cable. His published speech contains a date problem: it calls the shore arrival “Saturday, the 7th of September.” September 7, 1866, was a Friday. This edition retains that inconsistency in its note and uses the less precise “early September” in the narrative rather than silently choosing one half of the source's statement. [41] [42] [43]

Chapter 7 — What success meant

By the November banquet, Field could tell his audience that two cables were working. He contrasted an expectation of five words per minute with a claimed ability to send fifteen, and said the tariff had been cut in half. He hoped for a further reduction. These are valuable contemporary statements about capacity and price policy, but they are not a modern performance benchmark or an audited tariff series. The speech documents the claims and ambitions of a promoter explaining the new service to the commercial community that was honoring him. [43]

The same speech made a less triumphant admission. Field said there had been delays in receiving European messages, which he attributed to the land lines and the Gulf of St. Lawrence rather than the ocean cables. He described the frustration of a message crossing the Atlantic quickly and then waiting twenty-four hours to cross eighty miles of water. The precise allocation of blame belongs to Field; it has not been independently established here. The broader problem is nevertheless visible in both his admission and the newspaper's contemporary report of Gulf repair work. [44] [37]

Field also set out a standard that went beyond successful laying. The cables had to be kept in order, the connecting lines maintained, and a message sent from New York to London had to reach its destination promptly and without uncertainty. He called this an aim to secure, not merely a spectacle already accomplished. The aspiration supplies a useful ending for the narrative because it makes the user of the service visible. A sender wanted delivery, not only proof that a submarine conductor could carry a signal under favorable circumstances. [2]

> **Interpretation.** The difference between 1858 and 1866 is therefore larger than the difference between a first attempt and a later success. In 1858 an installed connection carried messages but could not be opened as the promised dependable service. In 1866 the sources describe a new working cable, the recovery and completion of another, and attention shifting toward traffic, rates, and connecting lines. This book stops within that transition. The evidence establishes the achievement by the end of 1866; it does not, on its own, measure how reliability or access changed over subsequent decades. [17] [18] [38] [43] [2]

The voices preserved here do not represent everybody affected by the undertaking. Russell's account privileges the expedition and its officers; Field's speech addresses merchants and bankers celebrating his success. The quoted royal message speaks of friendship between states, not of equal access for everyone who might wish to send a

message. This selection does not recover the experience of ordinary cable workers, Newfoundland households, or the communities that supplied materials. Their absence is a limit of this book's evidence, not evidence that they had no history worth telling. [27] [16] [21]

Nor can the language of international peace be taken as an outcome proved by the cable. Victoria's message expresses a hope. Field's account of pricing expresses an intention to make the service less expensive. Those statements are important objects of history even where their fulfillment has not been demonstrated. To repeat them as accomplished universal benefits would turn the aspirations of the sources into conclusions of the author. A narrower account can be both more cautious and more appreciative: the work made new communication possible without settling what people would do with it. [16] [43]

> **Interpretation.** The lasting interest of these attempts lies in the combination of ambition with correction. The project needed people willing to return after a loss, but it also needed alterations in machinery, changed material specifications, procedures for testing, a new financing arrangement, and methods for retrieving what had gone wrong. Persistence became useful through those particulars. “Never give up” is too thin a lesson for a history in which there were sound reasons to stop, inspect, postpone, and refuse to call a partial demonstration a finished service. [10] [11] [22] [24] [34] [39]

Interpretation. At the end, the line that could hold was more than the cable resting on the seabed. It was the continuing arrangement that connected a successful crossing to an actual delivered message. The small delay with which this book opened belongs beside the great arrival, not outside its history. It reminds us to honor the scale of the achievement while keeping its unfinished obligations in view. By 1866 the ocean connection had been made and renewed; the task of making dependable communication from it was work that success itself required. [1] [2]

Source note — What remains open

This narrative's central chronology is supported by the retained sources, but several questions remain deliberately unresolved. The full inquiry into the 1858 failure and the original traffic registers have not been examined. The account of that failure therefore reports Russell's summary rather than determining individual responsibility. The message count is a published count, not a fresh audit. Nor has this book measured the long-term effects on prices, trade, diplomacy, or inequality. Those larger claims would need a different body of evidence.

The 1866 landing date comes through Henry Field's reproduction of Cyrus Field's telegram; it is not the original dispatch sheet. The Newfoundland newspaper adds contemporary reporting of activity and messages, but some of that reporting also derives from Field. These sources are not four independent observers of every event. The day-and-date conflict in the recovery speech remains recorded, and the narrative deliberately avoids a false exactness there. Original facsimiles retained with this edition let readers examine the selected printed evidence themselves.

Digital transcription introduces another layer. Russell's text is a Project Gutenberg transcription of a reprint of his 1865 book. The banquet text is Internet Archive OCR, with errors and split words left visible in the quoted evidence. The short Henry Field and newspaper excerpts were transcribed from retained page images, with line-end hyphens joined and typographic punctuation normalized. A matching quotation proves only that this edition cites its saved text accurately. It does not prove that an OCR reading is correct or that the historical speaker's account was true.

Notes and exact excerpts

Quotations match the saved source text, including OCR errors and inserted page markers. A match establishes citation integrity, not historical truth. Locations refer to the printed work unless labeled “scan”.

1. Henry M. Field, p. 361.

   > Although the expedition reached Newfoundland on Friday, the twenty-seventh, yet, as the cable across the Gulf of St. Lawrence was broken, the news was not received in New-York till the twenty-ninth. It was early Sunday morning, before the Sabbath bells had rung their call to prayer, that the tidings came. The first announcement was brief: “Heart's Content, July 27.—We arrived here at nine o'clock this morning. All well. Thank God, the cable is laid, and is in perfect working order. Cyrus W. Field.”

   Henry Field’s reproduction of Cyrus Field’s telegram; family relationship and mediated transmission limit independence.

   Snapshot field-landing.txt · line 4 · Unicode code points [248, 751).

2. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 32.

   > To this end it must be kept in perfect order, and all lines connected with it. The very idea of an electric telegraph is, an instrument to send messages instantane ously. When a despatch is sent from New- York to Lon don, there must be no uncertainty about its reaching its destination — and that promptly. This we aim to secure.

   Snapshot banquet.txt · line 341 · Unicode code points [52846, 53175).

3. William Howard Russell, p. 6.

> Meantime the Company became involved in pecuniary difficulties, and Mr. Gisborne, early in 1854, on the suspension of the works, proceeded to New York, where he hoped to find money to enable him to carry out the telegraphic scheme among the keen speculators and large-pursed merchants. Through an accidental conversation at the hotel in which he was staying, he obtained an interview with Mr. Cyrus Field. He laid his plans before that gentleman

Snapshot russell.txt · line 41 · Unicode code points [13644, 14089).

4. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 19.

> It was easy to draw a line from one point to the other — making no account of the forests and mountains, and swamps and rivers and gulfs, that lay in our way. Not one of us had ever seen the country, or had any idea of the obstacles to be overcome. We thought we could build the line in a few months. It took two years and a half.

Snapshot banquet.txt · line 253 · Unicode code points [28315, 28645).

5. William Howard Russell, p. 8.

> After some little delay, the Legislature, with one adverse member only, granted the valuable privileges to the Company which were subsequently transferred to the Atlantic Telegraph Company. They constitute, in fact, a monopoly of telegraphic rights in Newfoundland

Snapshot russell.txt · line 57 · Unicode code points [20584, 20848).

6. William Howard Russell, p. 10.

> In August, 1855, the first attempt was made; but off Cape Ray a violent gale arose, and it was deemed necessary by the master of the vessel to cut the cable. This disappointment was not in the least a discouragement. Another contract was made by Mr. Field with Messrs. Küper & Co. to make and lay a cable at their own risk, which was executed by Mr. Canning in the Propontis the following year.

Snapshot russell.txt · line 83 · Unicode code points [26367, 26761).

7. William Howard Russell, p. 11.

> Mr. Field, in order to ascertain this fact, obtained from the government of America the assistance of Lieut. Berryman, U.S.N., in the steam-ship Arctic, who succeeded, in July, 1856, in taking soundings across the Atlantic at distances varying from 30 to 50 miles, and, by means of scoops, or quills, bringing up specimens of the bottom, which, upon microscopic examination, proved to be composed of fine shells and sand.

Snapshot russell.txt · line 93 · Unicode code points [28814, 29235).

8. William Howard Russell, p. 12.

> on the 6th December, 1856, they entered upon agreements with the Gutta Percha Company for the supply of 2,500 miles of core, consisting of copper wire, with a triple covering of insulating substance, at 40l. per mile; and also with Messrs. Glass, Elliot, & Co., of East Greenwich, and Messrs. Newall & Co., of Birkenhead, respectively, for the supply from each of 1,250 miles of the completed Cable for 62,000l. Within six months from that day, namely, on the 6th of July, 1857, the entire Cable was completed.

Snapshot russell.txt · line 109 · Unicode code points [32559, 33069).

9. William Howard Russell, p. 22.

> On the evening of Tuesday, the 11th, all signals suddenly ceased. The Cable had broken in 2000 fathoms of water, when about 330 nautical miles were laid, at a distance of 280 miles from Valentia. At the time the ship was going from three to four knots, and was able to pay out 5 to 5¾ miles per hour, the pressure shown by the indicator being 3000lb., but the strain being no doubt much greater.

Snapshot russell.txt · line 213 · Unicode code points [60420, 60815).

10. William Howard Russell, p. 22.

> This loss proved fatal to the first attempt to lay the Atlantic Cable, as on consultation among the officers and engineers it appeared to be unwise to renew the attempt with only 1,847 miles on board the ships, or an excess of 12 per cent. on the quantity required by the whole distance.

Snapshot russell.txt · line 215 · Unicode code points [60817, 61104).

11. William Howard Russell, p. 23.

> The brake of the paying-out machine used on the occasion of the first attempt was capable, by a movement of the hand, of exerting prodigious resistance. In the new machine any one could in a moment ease it, until there was no resistance at all beyond the 8 cwt. strain on the wire.
>
> At a few feet from the paying-out machine, the Cable passed over a wheel, which registered precisely the strain in pounds at which the coil was running out. Facing this register was a steering wheel, similar to that of an ordinary vessel, and connected in the same way with compound levers, which acted upon the brake. The officer in charge of the apparatus stood by this wheel, and watched the register of strain or pitch of the vessel, opening the brakes by the slightest movement of his hand, and letting the Cable run freely as the stern rose. The same officer, however, could not by any possible method increase the actual strain on the Cable, which remained always according to the friction at which the brake was at first adjusted by the engineer.

Snapshot russell.txt · line 227 · Unicode code points [64449, 65486).

12. William Howard Russell, pp. 23–24.

> Instead of landing a shore-end
>
> [Printed page 24]
>
> at Valentia, and making a junction of the Cable, it was decided that the ships should proceed together to a point midway between Trinity Bay and Valentia, there splice the Cable, and then turn their bows east and west, and proceed to their destinations.

Snapshot russell.txt · line 231 · Unicode code points [65619, 65923).

13. William Howard Russell, p. 25.

> At midnight on the 28th of July, 1858, the Agamemnon and Niagara once more met in mid-ocean, and on the following morning spliced the Cable, which was this time destined to tend so much towards solving the great problem. On the 30th, 265 miles had been paid out. On the 31st, 540 miles. On the 1st August, 884 miles. On the 2nd, 1256 miles. On the 4th, 1854 miles; and on the 5th, 2022 miles. The Agamemnon now anchored in Dowlas Bay, Valentia, and preparations were made to join the ocean and shore ends. On the same day, at 1·45 a.m., the Niagara anchored in Trinity Bay, Newfoundland, and in an hour after she received a signal across the Atlantic that the Cable had been landed from the Agamemnon.

Snapshot russell.txt · line 253 · Unicode code points [69918, 70619).

14. William Howard Russell, p. 24.

> On Thursday, the 10th of June, 1858, H.M.S. Agamemnon and U.S.N.S. Niagara, accompanied by H.M.S. Valorous and H.M.S. Gorgon, left Plymouth, the two former having previously made an experimental cruise in the Channel with the Cables, which were very satisfactory, in all respects.
>
> Experienced mariners gazed with apprehension at their depth in water as they left the shore. It was, however, such glorious weather as to cause some anxiety lest there should be no wind, and that the stock of coals might be exhausted before their mission was accomplished. Before midnight, however, a gradually increasing gale gathered to a storm, while the barometer marked 29°. For seven consecutive days the tempest, so eloquently described by Mr. Woods in the Times, continued, the Agamemnon under close-reefed topsails striving to reach the rendezvous, Lat. 52° 2´, Long. 33° 18´, rolling 45 degrees, and labouring fearfully.

Snapshot russell.txt · line 237 · Unicode code points [65925, 66837).

15. William Howard Russell, pp. 24–25.

> At half-past two o’clock on the 26th, the Agamemnon and Niagara first spliced the Cable; it however became foul of the scraper on the latter ship, and broke. A second splice was immediately made, and the vessels started. The Agamemnon had paid out 37½ miles, when suddenly the continuity of the electric current ceased, and the electricians declared that the Cable had broken at the bottom. As the Niagara was hauling in the Cable, of which she had payed out 43 miles, it snapped close to the ship.
>
> [Printed page 25]
>
> On the 28th, the third and final splice was effected. The Niagara started N.W. ¾ N. At 4 p.m. on the 29th, when 111 miles had been paid out, the electricians on board reported that continuity had ceased. The cause was soon known. The Agamemnon had run 118 miles, and paid out 146 miles of Cable, when the upper deck coil became exhausted. Speed was slackened, in order to shift the Cable to the lower deck, when suddenly it snapped, without any perceptible cause, under a strain of only 2200 pounds. The weather was calm; the speed moderate—about five knots; the strain one-third less than breaking strain; everything favourable; and yet the Cable parted, silently and suddenly.

Snapshot russell.txt · line 245 · Unicode code points [67796, 68993).

16. William Howard Russell, p. 26.

> The Queen desires to congratulate the President upon the successful completion of this great international work, in which the Queen has taken the deepest interest.
>
> “The Queen is convinced that the President will join with her in fervently hoping that the Electric Cable which now connects Great Britain with the United States will prove an additional link between the nations whose friendship is founded upon their common interest and reciprocal esteem.
>
> “The Queen has much pleasure in communicating with the President, and renewing to him her wishes for the prosperity of the United States.”

Snapshot russell.txt · line 275 · Unicode code points [72225, 72819).

17. William Howard Russell, p. 102.

> That messages can be transmitted through a Cable so laid, because 271 messages were sent from Newfoundland to Valentia, and 129 messages from Valentia to Newfoundland, in 1858.

Russell and Cyrus Field give the same total; this is not independent corroboration of the underlying traffic logs.

Snapshot russell.txt · line 1265 · Unicode code points [294929, 295105).

18. William Howard Russell, pp. 27–28.

> “SIR,—I am instructed by the Directors to inform you that, owing to some cause not at present ascertained, but believed to arise from a fault existing in the Cable at a point hitherto undiscovered, there have been no intelligible signals from Newfoundland since one o’clock on Friday the 3rd inst. The Directors are now in Valentia, and, aided by various scientific and practical electricians, are
>
> [Printed page 28]
>
> investigating the cause of the stoppage, with a view to remedying the existing difficulty. Under these circumstances no time can be named at present for opening the wire to the public.
>
> “GEO. SAWARD.”

Letter dated 6 September 1858, reproduced by Russell; the excerpt crosses a printed page boundary.

Snapshot russell.txt · line 327 · Unicode code points [75591, 76209).

19. William Howard Russell, p. 28.

> But this was of course mere conjecture, and the failure most probably was precipitated by injudicious attempts to overcome defective insulation by increased battery power.
>
> The conclusions finally arrived at by the Scientific Committee appointed to report as to the causes of the failure of the Cable were, first, that it had been manufactured too hastily; secondly, that a great and unequal strain was brought on it by the machinery; and thirdly, that the repeated coilings and uncoilings it underwent served to injure it. To such causes was the failure to be attributed, not to any original defect in the gutta percha.

Snapshot russell.txt · line 337 · Unicode code points [77623, 78243).

20. William Howard Russell, p. 29.

> In the month of April, 1860, the Directors of the Atlantic Telegraph Company sent out Captain Kell and Mr. Varley to Newfoundland to endeavour to recover some portion of the Cable; their efforts showed that the survey which had been taken must have been very insufficient, and the ground was much worse than was expected. They recovered five miles of the Cable, and ascertained two facts, namely, that the gutta percha was in no degree deteriorated, and that the electrical condition of the core had been improved by three years' submersion. In 1862 several attempts were also made to recover some of the Cable from the Irish side, but with no practical advantage; and in consequence of violent storms the attempt was abandoned.

Snapshot russell.txt · line 349 · Unicode code points [78935, 79663).

21. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 23.

> We needed £600,000, and with our utmost efforts we had raised less than half, and there the enterprise stood in a dead lock. It was plain that we must have help from some new quarter.

Snapshot banquet.txt · line 285 · Unicode code points [36864, 37047).

22. William Howard Russell, p. 32.

> The weight of the new conductor was nearly three times that of the former one—being 300 pounds to the nautical mile against 107 pounds per knot to the conductor of 1857.

Snapshot russell.txt · line 403 · Unicode code points [88150, 88319).

23. William Howard Russell, p. 33.

> The total weight of insulating material thus applied was 400 pounds to the nautical mile, against 261 pounds in the Cable of 1857-8.

Snapshot russell.txt · line 411 · Unicode code points [91669, 91801).

24. William Howard Russell, p. 33.

> The core, completed as described, and which had previously and repeatedly been under electrical examination, was at length submerged in water of a temperature of 75 deg. Fah., and so remained during twenty-four hours. This was done that the subsequent electrical tests for conductivity and insulation might be made under circumstances the most unfavourable to the manufacture, from the well-known fact, that the insulating power of gutta percha is sensibly decreased by heat. It also ensures uniformity of condition to the core under test,

Manufacturing description reproduced by Russell; a specification and account of testing, not an independently repeated experiment.

Snapshot russell.txt · line 413 · Unicode code points [91803, 92342).

25. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 25.

> As fast as ready, it was taken on board the " Great Eastern," and coiled in three enormous tanks; and on the 15th of July, 1865, the ship started on her memorable voyage.

Snapshot banquet.txt · line 301 · Unicode code points [40670, 40840).

26. William Howard Russell, p. 77; diary entry for 2 August 1865.

> The part of the Cable which had suffered from chafing was coming in, and the first portion of it was inboard; suddenly a jar was given to the dynamometer by a jerk, caused either by a heave of the vessel or by the shackle of wire-rope secured to the Cable, and the index jumped far above 60 cwt., the highest point marked on it. The chain shackle and wire-rope clambered up out of the groove of the V wheel, got on the rim, and rushed down with a crash on the smaller wheel, giving a severe shock to the Cable. Almost at the same moment, as the Cable and the rope travelled slowly along through the machinery, just ere they reached the dynamometer the Cable parted, flew through the stoppers, and with one bound leaped over intervening space and flashed into the sea.

Russell witnessed the expedition; his account leaves the precise initiating jerk uncertain.

Snapshot russell.txt · line 1051 · Unicode code points [220955, 221722).

27. William Howard Russell, p. 51.

> On board there were two representative bodies—the electricians of the Telegraph Construction and Maintenance Company, under M. de Sauty, and the electricians of the Atlantic Telegraph Company, Mr. Varley, Professor Thomson, and assistants. The former were to test the electrical state of the Cable as it was being payed-out, and to keep up signals between the ship and the shore. The latter, who had no power of interference or control, were simply to report on the testing, and to certify, on their arrival in Newfoundland, whether the Cable fulfilled the conditions specified in the contract.

Snapshot russell.txt · line 671 · Unicode code points [144141, 144735).

28. William Howard Russell, p. 74.

> Professor Thomson has since given it as his opinion that the fault could have been worked through, and that the Cable could have transmitted messages for a long time at the rate of four words a minute—making an amply remunerative return. Mr. de Sauty also entertained the belief that the Cable could have worked for several months, at all events. But it does not appear that Mr. Canning had any reason to act on the views of these gentlemen, and it was quite sure, when the end was landed in Heart’s Content, Mr. Varley could not have given his certificate that the Cable was of the contract standard.

Snapshot russell.txt · line 1031 · Unicode code points [212473, 213074).

29. William Howard Russell, p. 59.

> At 9 a.m., ship’s time, when somewhat more than 10¼ miles had been hauled in, to the joy of all the “fault” was discovered. The Cable came in with flagrant evidence of the mischief. The cause of so much anxiety, delay, and bitter disappointment turned out to be a piece of wire of the same kind as that used in the protecting strands of the Cable itself. It was two inches long or so—rather bent in the middle, with one end sharp and bright, as if from a sharp fracture or being cut by a pair of pliers—the other end blunt and jagged. This piece of wire had been forced through the outer covering of the Cable into the gutta percha, so as to injure the insulation, but no one could tell how it got into the tank. The general impression was, that it was a piece of Cable or other wire which had been accidently carried into the tank, and forced into the coil by the pressure of the paying-out machinery as the Cable flew between the jockey-wheels.

Snapshot russell.txt · line 891 · Unicode code points [167563, 168509).

30. William Howard Russell, p. 75.

> After all, the Cable might carry the source of deadly mischief within itself. What we had taken for assassination might have been suicide. The piece of wire in this case was evidently bad and brittle, and had started through the Manilla in the tank. How many similar pieces might have broken without being detected or causing loss of insulation? The marks of design in the second fault were very striking; but the freaks of machinery in motion are extraordinary, and what looked so like purposed malice might, after all, be the effect of accidental mechanical agency. There were thenceforth for the day two parties in the ship—those who believed in malice, and those who attributed all our disasters to accident. In the end the latter school included nearly all on board the ship, and it was generally thought that in the Cable, or, rather, in what had been intended as its protection, was the source of its weakness and ruin.

Snapshot russell.txt · line 1037 · Unicode code points [214342, 215268).

31. William Howard Russell, p. 102.

> That in a depth of two miles a Cable can be caught at the bottom, because four attempts were made to grapple the Cable in 1865, and in three of them the Cable was caught by the grapnel.

Snapshot russell.txt · line 1265 · Unicode code points [295854, 296039).

32. William Howard Russell, p. 115.

> Under this contract the Telegraph Construction and Maintenance Company undertake for the sum of 500,000l., which has been agreed on as the cost price, at once to commence the manufacture of and during 1866 to lay down, a new Cable between Ireland and Newfoundland.
>
> The Contractors, if the said Cable be successful, but not otherwise, are to have, in shares and cash, a profit at the rate of 20 per cent. upon such cost.
>
> The Contractors also undertake, without any further charge whatever, to go to sea with sufficient Cable, including that now left on board the Great Eastern, and all proper appliances and apparatus such as experience has shown to be necessary, and to use their best endeavours—in the success of which they entirely believe—to recover and repair and complete in working order between Ireland and Newfoundland, the present broken Cable.

Snapshot russell.txt · line 1935 · Unicode code points [316693, 317548).

33. William Howard Russell, p. 103.

> The Great Eastern is now undergoing the alterations which will render her absolutely perfect for the purpose of laying the new Cable and picking up the old,

This is Russell’s prediction in 1865, not proof that the next expedition was risk-free.

Snapshot russell.txt · line 1281 · Unicode code points [298441, 298597).

34. Chamber of Commerce of New-York; speech by Cyrus W. Field, pp. 26–27.

> The Attorney-General of England had given his written opinion that we had no legal right, without a special act of Parliament, (which could not be ob tained under a year,) to issue the new 12 per cent, shares, on which we relied to raise our capital. This was a ter rible blow. The works were at once stopped, and the money which had been paid in returned to the subscribers. Such was the state of things only ten months ago. I reached London on the 24th of December, and the next day was not a " merry Christmas " to me. But it was an inexpressible comfort to have the counsel of such men as Sir DANIEL GOOCH and Sir RICHARD A. GLASS ; and to hear stout-hearted Mr. BRASSEY tell us to go ahead ; and if need were, he would put down £60.000 more ! It was finally concluded that the best course was to organize a new company, which should assume the work ; and so
>
> 27
>
> originated the ANGLO-AMERICAN TELEGRAPH COMPANY.

Snapshot banquet.txt · line 307 · Unicode code points [41866, 42782).

35. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 27.

> Soon after, the books were opened to the public, through the eminent banking house of J. S. MORGAN & Co., and in fourteen days we had raised the whole £600,000.

Snapshot banquet.txt · line 311 · Unicode code points [43221, 43381).

36. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 27.

> Yet this was not ua lucky hit" — a fine run across the ocean in calm weather. It was the worst weather I ever knew at that season of the year. In the despatch which appeared in the New- York papers you may have read, " The weather has been most pleasant." I wrote it " un pleasant." We had fogs and storms almost the whole way. Our success was the result of the highest science combined with practical experience.

Snapshot banquet.txt · line 313 · Unicode code points [43899, 44312).

37. The Times and General Commercial Gazette (St. John’s, Newfoundland), scan p. 3, column 2, “TELEGRAMS”, Heart’s Content, 29 July.

> The Bloodhound arrived here at five this morning. She left again at three, with new Cable on board for the repair of the Gulf line.
> The Osprey arrived here at three p.m. to-day.
> The Great Eastern will remain here this week, and then proceed to pick up the old Atlantic Cable.
> Mr. Mackay left here on Friday night in H. M. S. Niger for Cape Breton with despatches from the Queen to the President.
> He will also superintend the repairing of the Gulf Cable.

Snapshot times.txt · line 6 · Unicode code points [401, 854).

38. The Times and General Commercial Gazette (St. John’s, Newfoundland), scan p. 3, column 2, “Postscript”, telegram of 31 July.

> Queen's message has been delivered to the President. Reply has been sent through the Cable to the Queen. Great Eastern is now firing a salute.
> CYRUS W. FIELD.”

Snapshot times.txt · line 15 · Unicode code points [1035, 1194).

39. Chamber of Commerce of New-York; speech by Cyrus W. Field, pp. 28–29.

> After finding it, they marked the line of the cable by a row of buoys ; for
>
> 29
>
> fogs would come down, and shut out sun and stars, so that no man could take an observation. These buoys were anchored a few miles apart. They were numbered, and each had a flag-staff on it, so that it could be seen by day ; and a lantern by night. Thus having taken our bearings, we stood off three or four miles, so as to come broadside on, and then casting over the grapnel, drifted slowly down upon it, dragging the bottom of the ocean as we went.

Snapshot banquet.txt · line 319 · Unicode code points [46368, 46899).

40. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 29.

> Once, on the 17th of , August, we got the cable up and had it in full sight for five minutes, a long, slimy monster, fresh from the ooze of the ocean's bed ; but our men began to cheer so wildly, that it seemed to be frightened, and suddenly broke away and went down into the sea. This accident kept us at work two weeks longer, but finally, on the last night of August, we caught it. We had cast the grapnel thirty times. It was a little before midnight on Friday night that we hooked the cable, and it was a little after midnight, Sun day morning, when we got it on board.

Snapshot banquet.txt · line 323 · Unicode code points [47560, 48134).

41. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 30.

> Then we carried it along to the electricians' room, to see if our long sought treasure was alive or dead. A few minutes of sus pense, and a flash told of the lightning current again set free.

Snapshot banquet.txt · line 327 · Unicode code points [48922, 49113).

42. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 30.

> and so, on Saturday, the 7th of September, we brought our second cable safely to the shore.

The facsimile also reads “Saturday, the 7th”. 7 September 1866 was Friday. This edition deliberately gives only “early September” for the shore landing; it does not silently repair this inconsistency.

Snapshot banquet.txt · line 327 · Unicode code points [50167, 50258).

43. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 35.

> Now these doubts are resolved. We have, not only one cable, but two, both in working order ; and we find, instead of five words a minute, we can send fifteen. Now we are free to reduce the tariff. Accordingly it has been cut down one-half, and I hope in a few months we can bring it down to one- quarter.

Snapshot banquet.txt · line 359 · Unicode code points [59237, 59541).

44. Chamber of Commerce of New-York; speech by Cyrus W. Field, p. 33.

> Yet there have been delays in re ceiving messages from Europe, but these have all been on the land lines or in the Gulf of St. Lawrence, and not on the sea cables. It was very painful to me, when we landed at Heart's Content, to find any interruption here, that a message which came in a flash across the Atlantic, should be delayed twenty-four hours in crossing eighty miles of water.

Snapshot banquet.txt · line 345 · Unicode code points [54539, 54924).

Bibliography and provenance

The Atlantic Telegraph (1865)

William Howard Russell · 1865.

Project Gutenberg no. 40948; transcription of a Naval Institute Press reprint of the 1865 work. The reprint front matter is retained in the raw HTML.

Eyewitness narrative for the 1865 voyage; retrospective account of 1854–1858; includes company and technical documents.

scripts/book_sources.py extracts text from printed page 1 onward, preserving printed-page markers and collapsing whitespace. Image captions are retained, images are not included.

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938091c4bc91a69128f3a53bf11a5e5e6ca527306b780a09fdbcc23258e0ec8f

The Atlantic Telegraph: Report of the Proceedings at a Banquet Given to Mr. Cyrus W. Field

Chamber of Commerce of New-York; speech by Cyrus W. Field · 1866-11-15.

New York: John W. Amerman, printer. Banquet at the Metropolitan Hotel, 15 November 1866. Internet Archive scan and OCR.

Participant speech after the successful expeditions. Celebration and self-justification are part of the source; it is not an independent engineering audit.

scripts/book_sources.py collapses whitespace in the retained OCR. OCR spelling and split words are not silently corrected; compare the retained PDF, printed page N = PDF page N+6 for the speech (e.g. p.30 = PDF page36).

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9256c050faed3b57d75ec4a9f3b3d2e8643cec85b9d232ec425ffa4a19cb54bb

History of the Atlantic Telegraph, selected passage

Henry M. Field · 1866.

1866 volume, printed p.361; Internet Archive / Google scan. Only the cited page is retained as a PDF shard.

Contemporary history by Cyrus Field’s brother, reproducing Cyrus’s landing telegram. This is a mediated report, not the original telegram or an independent witness.

Manual transcription of the final paragraph on p.361; line-end hyphens joined, quotation marks and dash spacing normalized. Ends at the page boundary. Facsimile shard made with PyMuPDF 1.26.5: source PDF page index363, zero-based.

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a80baa5c5171c34fb9c8eb673ceccc99cc0f2fccd7491b3ee96f00c9db3b115a

The Times and General Commercial Gazette: Telegrams and Postscript

The Times and General Commercial Gazette (St. John’s, Newfoundland) · 1866-08-01.

Memorial University Digital Archives Initiative scan; scan page3, second full column. Items dated Heart’s Content, July29 and July31.

Local newspaper publication close to the event; the Postscript is attributed to Cyrus Field and is not independent authorship of his telegram.

Manual selected-item transcription, not full OCR. Line-end hyphens joined, quotation marks/dash spacing normalized; italic emphasis omitted. Facsimile shard made with PyMuPDF 1.26.5: source PDF page index2, zero-based.

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